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Workpackage WPC

Implementation – Enterprise Characteristics

Deliverable C5: Quality template for statistical outputs

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Introduction

The structure of this quality template, named ESQRS 2.0 for online based enterprise characteristics (OBECs) is based on *Deliverable K6* produced by WPK (see on the following link: https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/8/87/WPK_Deliverable_K6_Quality_report_template_2020_02_28.pdf).

This template is derived from SIMS 2.0 for big data sources and intend to use for producer reports of experimental statistics. For the purpose it was elaborated a corresponding table by which only concepts regarding quality were extracted from SIMS and included in ESQRS, i.e. ESQRS 2.0 for OBECs comprises only concepts for the respective quality aspects for the experimental statistics on OBECs.

The definitions and guidelines are based on the recently published and updated version of the *ESS handbook for quality and metadata reports* (EHQMR)[1]. The new subconcepts for big data are indicated by an "A" for "additional".

Each country, which is to produce and publish experimental statistics for OBECs or other big data sources, can fill in this quality template to ensure that the quality is sufficient to use the data for disseminating experimental statistics.

In the Annex to this document, an example of ESQRS for OBECs that has been filled in Bulgaria is included. This example could help all potential usersof ESQRS 2.0 for OBECs.

Related documents:

Here you can find the Reference Methodological Framework where all quality aspects of OBECs data are described in details according to the work done by ESSnet Big Data I, WP-8 Methodology, ESSnet Big Data II, WP-K Quality Guidelines for the Usage of Big Data and UNECE Framework for the Quality of Big Data:
https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/index.php/WPC_Milestones_and_deliverables

[1] https://ec.europa.eu/eurostat/documents/3859598/10501168/KS-GQ-19-006-EN-N.pdf

Item №	Concept Name	Definition	Guidelines	OBECs
1	**Contact**	Individual or organisational contact points for the data or metadata, including information on how to reach the contact points.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
1.1	Contact organisation	The name of the organisation of the contact points for the data or metadata.	Provide the full name (not just code name). of organisation responsible for the process and outputs (data and metadata) that are the subject of the report.	
1.6	Contact email address	E-mail address of the contact points for the data or metadata.	Provide the email address(es) of the person(s) indicated as contacts. The address can be an individual e-mail address or a mailbox for the organisation to which the person has access.	-
1.7	Contact phone number	The telephone number of the contact points for the data or metadata.	Provide the telephone number(s) of the person(s) indicated as contacts.	
2	**Statistical presentation**	Description of the statistical output.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
2.1	Data description	Main characteristics of the data set, referring to the statistical output.	Describe briefly the main characteristics of the data in an easily and quickly understandable manner, referring to the main variables. More detailed descriptions of the variables and how they were derived in sub-concept 2.4.	
2.4	Statistical concepts and definitions	Statistical characteristics of statistical observations.	Define and describe briefly the main statistical variables that have been observed or derived. Indicate their types. Note that any difference between these variables and the variables desired by users is a relevance issue and is discussed in concept 5. Indicate discrepancies, if any, from variables which were previously collected in a different way (e.g. via surveys).	

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Item №	Concept Name	Definition	Guidelines	OBECs
2.5	Statistical unit	Entity for which information is sought and for which statistics are ultimately compiled.	Define the type of statistical unit about which data are available, e.g. enterprise, local unit, private household, person. If there is more than one type of unit, define each type. Summarize, if possible, the differences to units in traditional ways to collect data.	
2.6	Statistical population	The total membership or population or "universe" of a defined class of people, objects or events.	Define the target population of the statistical units for which information is sought. Note that a difference between the target population and the population desired by users is a relevance issue and is discussed in concept 5; and the difference between target population and the actual (frame) population is a coverage issue and is discussed in sub-concept 6.3. If there is more than one type of population, define each type. Describe if there are any differences to the populations in the traditional surveys. For example, the sub-population of OBEC enterprises will be always limited to enterprises having websites, filtered also by other attributes with respect to the population used in traditional survey.	
2.7	Reference area	The country or geographic area to which the measured statistical phenomenon relates.	Describe the country, the regions, the districts, or the other geographical aggregates, to which the data refer. Identify any specific exclusions in the statistical data.	
2.8	Time coverage	The length of time for which data are available.	State the time period(s) covered by the data, e.g. first quarter 2018, or quarters 2015-2018, or year 2018, or years 1985-2018. Note that any issues concerning comparability over time are discussed in concept 8.	
3	**Statistical processing**	(Defined by its sub-concepts)	(Information relating to this concept is provided by reporting on its sub-concepts.)	

Item №	Concept Name	Definition	Guidelines	OBECs
3.1	Source data	Characteristics and components of the raw statistical data used for compiling statistical aggregates.	Indicate if the data are based on a survey, an administrative data source, multiple data sources, big data source (machine generated, human sourced, process mediated), e.g., web data, and/or macro-aggregates. Refer to the accreditation document of the data source, if applies. In the event of multiple data sources or macro-aggregates, reference each source and indicate how they are combined. For each survey source, summarise the sample design, cross referencing the descriptions of the target and survey populations, presented in sub-concept 2.6. For each administrative data source, summarise the source, its primary purpose, and the most important data items acquired. Information in which form the metadata for the new data source is available, where it can be found, and if it is updated on a regular basis. **European level** Provide an overview of the sources used across countries.	
3.2	Frequency of data collection	Frequency with which the source data are collected.	Indicate the frequency of data collection (e.g. monthly, quarterly, annually, or continuous).	
3.3	Data collection	Systematic process of gathering data for official statistics.	For each survey data source: • describe the method(s) used to gather data from respondents; • annex or hyperlink the questionnaires(s). For each administrative data source	

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			• describe the acquisition process and how it was tested. For all sources • describe the types of checks applied at the time of data entry. For big data sources describe the methods used to collect the data; add hyperlink if it is web data or name of the API used to collect the data. **European level** Provide a summary of the commonalities and differences in the collection methods, questionnaires and checks used in different countries.	
3.4	Data validation	Process of monitoring the results of data compilation and ensuring the quality of statistical results.	Describe the procedures for checking and validating the source data and how the results are monitored and used. Describe the procedures for validating the aggregate output data (statistics) after compilation, including checking coverage and response rates, and comparing with data for previous cycles and with expectations. List other output datasets to which the data relate and outline the procedures for identifying inconsistencies between the output data and these other datasets. Define the linkage method for big data sources and other data sources used for validation. **European level** Provide a summary of the commonalities and differences in the validation methods used by countries.	

Item №	Concept Name	Definition	Guidelines	OBECs
3.5	Data compilation	Operations performed on data to derive new information according to a given set of rules.	If there is missing data, give detailed description of the methods used for imputation. For big data sources, e.g., web data, indicate the reason why data were not collected (technical issues etc.). Describe the procedures for imputation, the most common reasons for imputation and imputation rates within each of the main strata. Describe the likely impact of imputation. Describe the procedures for adjustment for non-response and the corrections to the design weights to account for differences in response rates. Describe the calculation of design weights, including calibration (if used). Describe the procedures for combining input data from different sources.	
4	**Quality management**	Systems and frameworks in place within an organisation to manage the quality of statistical products and processes.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
4.1	Quality assurance	All systematic activities implemented that can be demonstrated to provide confidence that the processes will fulfil the requirements for the statistical output.	Describe the quality assurance procedures specifically applied to the statistical process for which the report is being prepared, for example agreements with the big data providers, benchmarking, assessments, and use of best practices. Include descriptions of all forms of quality assessment procedures (self-assessment, peer review, compliance monitoring, audit) and when they most recently took place. Summarise the results of the most recent quality assessments and cross reference to the chapters in the	

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Item №	Concept Name	Definition	Guidelines	OBECs
			report where the results are presented in more detail. Describe any ongoing or planned improvements in quality assurance procedures.	
4.2	Quality assessment	Overall assessment of data quality, based on standard quality criteria.	Summarise the results of the most recent quality assessments and cross reference to the chapters in the report where the results are presented in more detail.	
5	**Relevance**	The degree to which statistical information meet current and potential needs of the users.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
5.1	User needs	Description of users and their respective needs with respect to the statistical data.	Provide: • a classification of users, also indicating their relative importance; • an indication of the uses for which users want the statistical outputs; • an assessment of the key outputs desired by different categories of users and any shortcomings in outputs for important users; • information on unmet user needs and any plans to satisfy them in the future; and • details regarding those quality components which do not meet user requirements.	

Item №	Concept Name	Definition	Guidelines	OBECs
5.3	Completeness	The extent to which all statistics that are needed are available.	Provide qualitative information on the extent to which content requirements in relevant legislation, regulations and guidelines are met. Provide information on the extent to which user needs related to content are satisfied. Provide values of indicator R1 Data completeness rate, for each required data item for each relevant regulation/ guideline at producer/user level of detail as appropriate. In the case where the indicator refers to data sent to Eurostat, this indicator can be compiled by Eurostat. European level Summarise across countries the extent to which ESS requirements for data items are met	
5.A	Added Value through new data source	The potential added value of a new data source to an existing statistical product.	Describe if and how the usage of a new data source provides an added value to an already existing statistical product. E.g., this could be more detailed data on particular subgroups, or information on grid level instead of district level or the potential replacement of questions of a survey through information of the new data source.	
6	**Accuracy and reliability**	Closeness of computations or estimates to the exact or true values that the statistics were intended to measure. Reliability of the data, defined as the closeness of the initial estimated value to the subsequent estimated value.	(Information relating to accuracy is provided by reporting on 6 sub-concepts. Information on Reliability is reported in sub-concept 6.5 Data Revision-policy).	

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Item №	Concept Name	Definition	Guidelines	OBECs
6.1	Overall accuracy	Assessment of accuracy, linked to a certain data set or domain, which is summarising the various components.	Describe the main sources of random and systematic errors in the statistical outputs and provide a summary assessment of all errors with special focus on the impact on key estimates. The bias assessment can be in quantitative or qualitative terms, or both, and may be expressed as bias risk. It should reflect the producer's best current understanding (sign and order of magnitude) and include actions taken to reduce bias. **European level** Provide a summary picture of accuracy across countries. The emphasis placed on various types of errors should depend upon the error profile of the respective process. For repetitive processes, describe how accuracy is developing over time and what efforts are underway to improve accuracy from an ESS perspective. There is a tendency to focus on the micro-level here. please include in this and subsequent sections that reporting at the group or aggregated level can/should be done when the units can not be identified. In general, one should be able to repport any quality issues when working with event-based Big Data sources!!!	

Item №	Concept Name	Definition	Guidelines	OBECs
6.2	Sampling error	That part of the difference between a population value and an estimate thereof, derived from a random sample, which is due to the fact that only a subset of the population is enumerated.	State whether sampling error is relevant. If probability sampling is used: • for user reports, provide the range of variation of the A1 indicator among key variables at user report level of detail; • for producer reports, provide the range of variation of the A1 indicator among key variables at producer report level of detail; • indicate the impact of sampling error on the overall accuracy of the results; • state how the calculation of sampling error is affected by imputation for nonresponse, misclassifications and other sources of uncertainty, such as outlier treatment. If non-probability sampling is used, provide an assessment of representativity and risk of sampling bias. **ESS level** If probability sampling is used: • present sampling errors for key estimates across countries; • indicate which country to country differences are significant and which are not; • for a repetitive survey, describe at least broadly the trends in sampling error over time; • provide sampling errors for ESS level estimates.	
6.3	Non-sampling error	Error in estimates which cannot be attributed to sampling fluctuations	Summarise the most important aspects of coverage, measurement, non-response, processing and model assumption errors. Discuss the corresponding bias risks and actions undertaken to reduce them.	

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Item №	Concept Name	Definition	Guidelines	OBECs
	A4. Unit non-response - rate (U)	The ratio of the number of units with no information or not usable information to the total number of in-scope (eligible) units, at a level of detail appropriate for a user report.		
	A5. Item non-response - rate (U)	The ratio of the in-scope (eligible) units that have not provided a particular item and the in-scope units that are expected to provide that particular item, at a level of detail appropriate for a user report.		
6.3.1	Coverage error	Divergence between the population of the Big Data source and the target population.	Provide information on the frame and its sources and actions performed to gather the population impacting on coverage (e.g. webscraping). Provide an assessment, whenever possible quantitative, of overcoverage and undercoverage, including an evaluation of the bias risks associated with the latter. Describe actions taken for reduction of undercoverage and associated bias risks	
6.3.1.1	A2. Overcoverage – rate (P)	The proportion of units accessible via the frame that do not belong to the target population.	Report A2, Overcoverage - rate	

Item №	Concept Name	Definition	Guidelines	OBECs
6.3.2	Measurement error	Measurement errors are errors that occur during data capture and cause recorded values of variables to be different from the true ones	The main sources of measurement error should be reported and assessed. Their description should be accompanied by any available analysis, otherwise by the producer's best knowledge. Where available and relevant describe: • identification and general assessment of the main sources of measurement error, including errors arising from data acquisition; • efforts made in questionnaire design and testing, information on interviewer training and other work on error prevention; • errors in measurement instruments (meters, satellites,...) • results of assessments based on comparisons with external data, re-interviews or experiments; • results of indirect analysis, for example, of the editing phase; and • actions taken to correct measurement errors.	
6.3.3	Non response error	Nonresponse errors occur when the Big Data source fails to collect one or all the variables for units belonging to the domain covered by the source	Provide qualitative/quantitative assessments of unit nonresponse and highlight the units that are most subject to nonresponse. Highlight the variables that are most subject to item nonresponse. Provide a qualitative/quantitative assessments of the bias associated with nonresponse, comparing response rate for different sub-groups or distribution of auxiliary variables known for respondents and non-respondets (etc.). Provide a breakdown of nonrespondents according to cause for nonresponse mainly focusing on unit dependent cause and data collection tools cause. Define a stategy for reducing nonresponse during	

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Item №	Concept Name	Definition	Guidelines	OBECs
			data collection and follow-up. Implement an estimator adjusted for nonresponse . **European level:** Provide a qualitative/quantitative assessments of unit and item nonresponse across countries.	
6.3.3.1	A4. Unit nonresponse - rate (P)	The ratio of the number of units with no information or not usable information to the total number of in-scope (eligible) units, at a level of detail appropriate for a producer report.	Report A4: Unit nonresponse rate overall and at a level of detail appropriate for a producer report.	
6.3.4	Processing error	The error in final data collection process results arising from the faulty implementation of correctly planned implementation methods, e.g., algorithms used to transform the data or extract information from raw data.	If processing errors are significant, identify the main issues regarding them. Present an analysis of processing errors, where available, otherwise a qualitative assessment. Report their extent, and impact on the outputs, of the most significant types of error. Include descriptions of linking and coding errors, if applicable. Where mistakes relating to programming or publishing have occurred, corrective measures taken as well as actions for avoiding them in the future should be reported. Example: For web data sources: Setting up a pipeline assures processing is comparable over time. Because texts were processed, the final results were highly affected by the various choices of text processing made.	

Item №	Concept Name	Definition	Guidelines	OBECs
6.3.4.1	A7. Imputation – rate	The ratio of the number of replaced values to the total number of values for a given variable.	Provide values of indicator A7 Imputation – rate	
6.3.5	Model assumption error	Error due to models used in the statistical production.	Describe process specific models, for example, as needed to define the target of estimation itself and models used for transformation of data into statistical data. Provide an assessment of the validity of each model. Descriptions of models used in treatment of specific sources of error should be presented in the section dealing with those errors. The assessment of the models used in treatment of specific sources of error should be presented in this section. Discuss the trade off between the need to use proper model that can change over time (accuracy) and the use a constant model in order to ensure comparability over time	
6.5	Data revision - policy	Policy aimed at ensuring the transparency of disseminated data, whereby preliminary data are compiled that are later revised.	Describe the data revision policy applicable to data output from the statistical process being reported. In so far as they are relevant to the process being reported, summarise the general procedures for treatment of planned revisions, benchmark revisions, unplanned revisions, and revisions due to conceptual and/or methodological changes. **European level** Describe the data revision policy and procedures at European level.	

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7	**Timeliness and punctuality**	(Defined by its sub-concepts)	(Information relating to this concept is provided by reporting on its sub-concepts.)	
7.1	Timeliness	Length of time between data availability and the event or phenomenon the data describe.	Outline the reasons for the time lag. Outline efforts to reduce time lag in future. Describe the envisioned time lag for producing statistical output from/with the help of a new data source. Describe if the use of the new data source has the potential to decrease the time lag as it exists at the moment for already existing statistical products.	
8	**Coherence and comparability**	Adequacy of statistics to be reliably combined in different ways and for various uses and the extent to which differences between statistics can be attributed to differences between the true values of the statistical characteristics.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
8.1	Comparability - geographical	The extent to which statistics are comparable between geographical areas.	Describe any problems of comparability between regions of the country. The reasons for the problems should be described and as well an assessment (preferably quantitative) of the possible effect on the output values. Give information on discrepancies from the ESS/ international concepts, definitions, with reference to other chapters for more details.	

Item №	Concept Name	Definition	Guidelines	OBECs
8.2	Comparability - over time	The extent to which statistics are comparable or reconcilable over time.	Provide information on possible limitations in the use of data for comparisons over time. Distinguish three broad possibilities: 1. There have been no changes, in which case this should be reported. 2. There have been some changes but not enough to warrant the designation of a break in series. 3. There have been sufficient changes to warrant the designation of a break in series. Additionally, provide information about the comparability over time of the technological processes which produce the data, of the data access and changes in the covered population over time. Give also an assessment how the the comparison over time will develop in the future.	
8.3	Coherence - cross domain	The extent to which statistics are reconcilable with those obtained through other data sources or statistical domains.	An analysis of incoherence should be provided, where this is an issue of importance.	
8.4	Coherence - internal	The extent to which statistics are consistent within a given data set.	Each set of outputs should be internally consistent. If statistical outputs within the data set in question are not consistent, any resulting lack of coherence in the output of the statistical process itself should be stated as well as a brief explanation of the reasons for publishing such results.	
8.A.1	Coherence - with existing information/ Official Statistics	The extent to which information / statistical output from new data sources is consistent with information /statistical output from traditional data sources.	Provide information if it is meaningful to compare the information gained from new data sources with information from traditional data sources and if so, how consistent the information /statistical output gained from new data sources is with the one from traditional data sources.	

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Item №	Concept Name	Definition	Guidelines	OBECs
8.A.2	Comparability - between information from several distinct new data sources	The extent to which information from several distinct new data sources is comparable among one another.	If you have raw data from several distinct new data sources, provide information how comparable the respective raw data sets and the information derived from them are among one other. Examples: MNO data from several mobile operators, smart meter data from several electricity providers	
9	**Accessibility and clarity**	The conditions and modalities by which users can access, use and interpret data.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
9.6	Documentation on methodology	Descriptive text and references to methodological documents available.	List national reference metadata files, methodological papers, summary documents and handbooks relevant to the statistical process. For each item provide the title, publisher, year and link to on-line version (if any). List deliverables, reference metadata files, methodological papers, summary documents etc relevant to the process of deriving statistical data from raw data and - if already available - for producing statistical output using the statistical data.	
9.7	Quality documentation	Documentation on procedures applied for quality management and quality assessment.	List relevant quality related documents, for example, other quality reports, studies. Cross reference to descriptions of quality procedures in other chapters, especially concept 6. List also the deliverables, in which quality related issues are described.	

Item №	Concept Name	Definition	Guidelines	OBECs
10	**Cost and burden**	Cost associated with the collection and production of a statistical product and burden on respondents.	Cost Provide annual operational costs of the process, with breakdown by major cost component. Describe recent efforts to improve efficiency and comment on the extent to which information and communication technology is used. European level Describe recent initiatives and efforts to improve efficiency at the European level. Burden Provide an estimate of the respondent burden imposed by the process. Describe all the means taken to minimise burden. European level Describe recent initiatives and efforts to minimise burden at the European level.	
10.A	Potential savings in cost and burden	Description how the new data source might influence cost and burden in the future	Provide an overview how the new data source could be deployed in the future to save the NSIs cost and/or decrease the respondent burden. Provide a qualitative description of the additional efforts for the NSI and the data owners.	
11	**Confidentiality**	A property of data indicating the extent to which their unauthorised disclosure could be prejudicial or harmful to the interest of the source or other relevant parties.	(Information relating to this concept is provided by reporting on its sub-concepts.)	

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Item №	Concept Name	Definition	Guidelines	OBECs
11.1	Confidentiality - policy	Legislative measures or other formal procedures which prevent unauthorised disclosure of data that identify a person or economic entity either directly or indirectly.	Describe all European or national legislation, or other formal requirements, that relate to confidentiality. Describe relevant policy (if any). Note that the existence of legislation and/or policy provides some assurance that methods necessary to assure confidentiality have been applied to the data. European level Summarise the commonalties and differences in national approaches to confidentiality policy	
11.2	Confidentiality - data treatment	Rules applied for treating the datasets to ensure statistical confidentiality and prevent unauthorised disclosure.	For aggregate outputs • Provide the rules that define a confidential cell. • Describe the procedures for detecting confidential cells, including checking for residual disclosure. • Describe the procedures for eliminating confidential cells, for example by controlled rounding, cell suppression, or cell aggregation. For micro-level outputs: • Describe the procedures that are used in protecting confidentiality.	
11.A1	Privacy	How privacy sensitive is the information coming from external data holders?	State which treatments are prescribed to satisfy privacy concerns	
11.A2	Privacy - protecting treatments	Treatment applied to ensure privacy - sensitive information from external data holders	State any treatments prescribed to satisfy privacy concerns.	
12	**Comment**	Supplementary descriptive text which can be attached to data or metadata.	Provide any information that is pertinent to the report but does not fit under any of the other concepts, or to repeat key issues, or to make reference to annexes that might be attached to the report.	

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1	**Contact**	Individual or organisational contact points for the data or metadata, including information on how to reach the contact points.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
1.1	Contact organisation	The name of the organisation of the contact points for the data or metadata.	Provide the full name (not just code name). of organisation responsible for the process and outputs (data and metadata) that are the subject of the report.	Bulgarian National Statistical Institute (BNSI)
1.6	Contact email address	E-mail address of the contact points for the data or metadata.	Provide the email address(es) of the person(s) indicated as contacts. The address can be an individual e-mail address or a mailbox for the organisation to which the person has access.	gstateva@nsi.bg kgeorgiev@nsi.bg
1.7	Contact phone number	The telephone number of the contact points for the data or metadata.	Provide the telephone number(s) of the person(s) indicated as contacts.	+359 2 9857773
2	**Statistical presentation**	Description of the statistical output.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
2.1	Data description	Main characteristics of the data set, referring to the statistical output.	Describe briefly the main characteristics of the data in an easily and quickly understandable manner, referring to the main variables. More detailed descriptions of the variables and how they were derived in sub-concept 2.4.	On-line characteristics of enterprises (OBECs) with 10 and more employees, scraped from their websites and Search engines, e.g. URLs, e-commerce, social media presence, job advertisments and etc.

2.4	Statistical concepts and definitions	Statistical characteristics of statistical observations.	Define and describe briefly the main statistical variables that have been observed or derived. Indicate their types. Note that any difference between these variables and the variables desired by users is a relevance issue and is discussed in concept 5. Indicate discrepancies, if any, from variables which were previously collected in a different way (e.g. via surveys).	The main statistical concepts covered are the following: • Enterprise • Website, home-page • E-commerce • Social Media Presence • Economic activity (NACE rev. 2) • Number of employed persons • Regional level by NUTS 3 The main statistical indicators are: 1. Rate of enterprises having websites a. Size of the enterprises according to number of persons employed; b. Enterprises by NACE categories; c. Enterprises by NUTS Level 3; 2. E-commerce - Rate of enterprises engaged in websales on their website 3. Social Media Presence - rate of enterprises that are present on social media 4. Rate of enterprises having specific features of the website: a. Advertisement of open job positions or online job application.
2.5	Statistical unit	Entity for which information is sought and for which statistics are ultimately compiled.	Define the type of statistical unit about which data are available, e.g. enterprise, local unit, private household, person. If there is more than one type of unit, define each type. Summarize, if possible, the differences to units in traditional ways to collect data.	Enterprise.

2.6	Statistical population	The total membership or population or "universe" of a defined class of people, objects or events.	Define the target population of the statistical units for which information is sought. Note that a difference between the target population and the population desired by users is a relevance issue and is discussed in concept 5; and the difference between target population and the actual (frame) population is a coverage issue and is discussed in sub-concept 6.3. If there is more than one type of population, define each type. Describe if there are any differences to the populations in the traditional surveys. For example, the sub-population of OBEC enterprises will be always limited to enterprises having websites, filtered also by other attributes with respect to the population used in traditional survey	The traditional survey consists of all enterprises limited by NACE having 10 or more employees. In contrast, the OBEC survey consists of the sub-population of enterprises limited by NACE and having 10 or more employees as well as having the website, based on URL stored in Business Register or identified by software dedicated to retrieve URLs for enterprises. Target population by *ICT usage in enterprises survey* according to the Regulation No.808/2004 of the European Parliament and the Council: enterprises by limited NACE categories and with 10 and more employees. This target population is extracted from SBR and it is used only for producing of the *Rate of enterprises having websites* indicator. The sub-population used for all other OBECs indicators includes only the enterprises with URLs from the target population.
2.7	Reference area	The country or geographic area to which the measured statistical phenomenon relates.	Describe the country, the regions, the districts, or the other geographical aggregates, to which the data refer. Identify any specific exclusions in the statistical data.	Bulgaria and Bulgarian NUTS 3 level.
2.8	Time coverage	The length of time for which data are available.	State the time period(s) covered by the data, e.g. first quarter 2018, or quarters 2015-2018, or year 2018, or years 1985-2018. Note that any issues concerning comparability over time are discussed in concept 8.	Year 2020, time series for the period 2017-2020.

3	**Statistical processing**	(Defined by its sub-concepts)	(Information relating to this concept is provided by reporting on its sub-concepts.)	

3.1	Source data	Characteristics and components of the raw statistical data used for compiling statistical aggregates.	Indicate if the data are based on a survey, an administrative data source, multiple data sources, big data source (machine generated, human sourced, process mediated), e.g., web data, and/or macro-aggregates.
			
			Refer to the accreditation document of the data source, if applies. In the event of multiple data sources or macro-aggregates, reference each source and indicate how they are combined. For each survey source, summarise the sample design, cross referencing the descriptions of the target and survey populations, presented in sub-concept 2.6. For each administrative data source, summarise the source, its primary purpose, and the most important data items acquired. Information in which form the metadata for the new data source is available, where it can be found, and if it is updated on a regular basis.
			European level
			Provide an overview of the sources used across countries.
				BNSI used the following data sources to produce experimental OBECs statistics:
				• Statistical Business Register (SBR)
				  • Enterprise ID
				  • Enterprise Name
				  • Enterprise E-mail
				  • Enterprise known web site
				  • Enterprise post address
				  • Enterprise NUTS 3 level
				  • Enterprise NACE information
				• DuckDuckGo Search Engine
				  • 10 suggestions for enterprise URL
				• Enterprises web sites
				  • Up to 20 pages from the enterprise web site
				• ICT usage in enterprises survey

3.2	Frequency of data collection	Frequency with which the source data are collected.	Indicate the frequency of data collection (e.g. monthly, quarterly, annually, or continuous).	Annually.

3.3	Data collection	Systematic process of gathering data for official statistics.	For each survey data source: • describe the method(s) used to gather data from respondents; • annex or hyperlink the questionnaires(s). For each administrative data source • describe the acquisition process and how it was tested. For all sources • describe the types of checks applied at the time of data entry. For big data sources describe the methods used to collect the data; add hyperlink if it is web data or name of the API used to collect the data. **European level** Provide a summary of the commonalities and differences in the collection methods, questionnaires and checks used in different countries.	A list of enterprise and their characteristics (target population for ICT usage in enterprises survey) is requested from the SBR. The name of enterprises are searched in DuckDuckGo search engine and up to 10 suggested URLs for each enterprise are stored. For each URL up to 20 pages are scraped and scored for known enterprise's characteristics. The scores are used for applying the ML methods for determination of real URLs of enterpises. The URLs of enterpises are used to scrape and score up to 20 pages for determination of OBECs by applying the ML techniques.
3.4	Data validation	Process of monitoring the results of data compilation and ensuring the quality of statistical results.	Describe the procedures for checking and validating the source data and how the results are monitored and used. Describe the procedures for validating the aggregate output data (statistics) after compilation, including checking coverage and response rates, and comparing with data for previous cycles and with expectations. List other output datasets to which the data relate and outline the procedures for identifying inconsistencies between the output data and these other datasets. Define the linkage method for big data	The results from ML procedure applied for URLs and OBECs are manually validated for those that are different from what we already known. The training and test set for ML procedure are splited as 70% to 30%. The ML procedure was repeated 1000 times for URLs Retrieval and 100 times for OBECs and the results were concatinated before the validation. The output data was validated by comparing the results with aggregated data reported by the

			sources and other data sources used for validation. **European level** Provide a summary of the commonalities and differences in the validation methods used by countries.	most recent ICT usage in enterprise survey results.
3.5	Data compilation	Operations performed on data to derive new information according to a given set of rules.	If there is missing data, give detailed description of the methods used for imputation. For big data sources, e.g., web data, indicate the reason why data were not collected (technical issues etc.). Describe the procedures for imputation, the most common reasons for imputation and imputation rates within each of the main strata. Describe the likely impact of imputation. Describe the procedures for adjustment for non-response and the corrections to the design weights to account for differences in response rates. Describe the calculation of design weights, including calibration (if used). Describe the procedures for combining input data from different sources.	Some web data are not collected because of the Internet connection problems categorized as follows: • HTTP error; • Time out accurred; • To many re-directs; • Request exceptions; • General exception. No blocking from the source websites was detected during the webscraping process.

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4	**Quality management**	Systems and frameworks in place within an organisation to manage the quality of statistical products and processes.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
4.1	Quality assurance	All systematic activities implemented that can be demonstrated to provide confidence that the processes will fulfil the requirements for the statistical output.	Describe the quality assurance procedures specifically applied to the statistical process for which the report is being prepared, for example agreements with the big data providers, benchmarking, assessments, and use of best practices. Include descriptions of all forms of quality assessment procedures (self-assessment, peer review, compliance monitoring, audit) and when they most recently took place. Summarise the results of the most recent quality assessments and cross reference to the chapters in the report where the results are presented in more detail. Describe any ongoing or planned improvements in quality assurance procedures.	The data are scrapped only ones (in the reference period) because the number of non-responsive websites were less than 0,5 % for OBECs and 0,3% for URLs. URLs are linked to enterprises and validated by Logistic Regression ML algorithm. The ML procedure was repeated 1000 times for URLs Retrieval and 100 times for OBECs and the results were concatinated before the validation. Then the deterministic approach is applied only for true negative, false positive and false negative results to determine the real URLs or OBECs data.
4.2	Quality assessment	Overall assessment of data quality, based on standard quality criteria.	Summarise the results of the most recent quality assessments and cross reference to the chapters in the report where the results are presented in more detail.	Validation of URLs and OBECs described on the Quality chapter, RMF 2.0.
5	**Relevance**	The degree to which statistical information meet current and potential needs of the users.	(Information relating to this concept is provided by reporting on its sub-concepts.)	

5.1	User needs	Description of users and their respective needs with respect to the statistical data.	Provide: • a classification of users, also indicating their relative importance; • an indication of the uses for which users want the statistical outputs; • an assessment of the key outputs desired by different categories of users and any shortcomings in outputs for important users; • information on unmet user needs and any plans to satisfy them in the future; and •details regarding those quality components which do not meet user requirements.	The main users are subject matter esperts from ICT usage in enterprise survey. They could use the experimental OBECs data as a complementary source to validate the respone by the respondents. In addition the SBR esperts could use the OBECs data for the updauing the SBR information for enterprises and to have a general population with higher quality for bussness statistical surveys. The general users who have an interest for OBECs data.
5.3	Completeness	The extent to which all statistics that are needed are available.	Provide qualitative information on the extent to which content requirements in relevant legislation, regulations and guidelines are met. Provide information on the extent to which user needs related to content are satisfied. Provide values of indicator R1 Data completeness rate, for each required data item for each relevant regulation/ guideline at producer/user level of detail as appropriate. In the case where the indicator refers to data sent to Eurostat, this indicator can be compiled by Eurostat. **European level** Summarise across countries the extent to which ESS requirements for data items are met	The statistical indicators in the RMF 2.0 which are not produced are as follows: • Rate of enterprises using Twitter for a specific purpose • Rate of enterprises working on upcoming/new phenomena, specifically AI and ML

5.A	Added Value through new data source	The potential added value of a new data source to an existing statistical product.	Describe if and how the usage of a new data source provides an added value to an already existing statistical product. E.g., this could be more detailed data on particular subgroups, or information on grid level instead of district level or the potential replacement of questions of a survey through information of the new data source.	New approach allows having enterprises even with less than 10 employees. The population can be bigger than in traditional survey. However there will be over-coverage of statistical units having websites and no data for enterprises not having websites. The data can also be produced at a much higher frequency and higher timeliness. The outputs could be distributed by NUTS 3 level and NACE Rev.2 categories. The results are not the time and cost consuming.
6	**Accuracy and reliability**	Closeness of computations or estimates to the exact or true values that the statistics were intended to measure. Reliability of the data, defined as the closeness of the initial estimated value to the subsequent estimated value.	(Information relating to accuracy is provided by reporting on 6 sub-concepts. Information on Reliability is reported in sub-concept 6.5 Data Revision-policy).	
6.1	Overall accuracy	Assessment of accuracy, linked to a certain data set or domain, which is summarising the various components.	Describe the main sources of random and systematic errors in the statistical outputs and provide a summary assessment of all errors with special focus on the impact on key estimates. The bias assessment can be in quantitative or qualitative terms, or both, and may be expressed as bias risk. It should reflect the producer's best current understanding (sign and order of magnitude) and include actions taken to reduce bias. **European level**	The main sources for errors are the following: • The search engine doesn't provide the enterprise URL in the first 10 suggestions; • The lists of key words for OBECs are not relevant enough; • The misleading content of websites; • Human errors through the deterministic approach; • Logistic Regression ML approach;

			Provide a summary picture of accuracy across countries. The emphasis placed on various types of errors should depend upon the error profile of the respective process. For repetitive processes, describe how accuracy is developing over time and what efforts are underway to improve accuracy from an ESS perspective. There is a tendency to focus on the micro-level here. please include in this and subsequent sections that reporting at the group or aggregated level can/should be done when the units can not be identified. In general, one should be able to repport any quality issues when working with event-based Big Data sources!!!	
6.2	Sampling error	That part of the difference between a population value and an estimate thereof, derived from a random sample, which is due to the fact that only a subset of the population is enumerated.	State whether sampling error is relevant. If probability sampling is used: • for user reports, provide the range of variation of the A1 indicator among key variables at user report level of detail; • for producer reports, provide the range of variation of the A1 indicator among key variables at producer report level of detail; • indicate the impact of sampling error on the overall accuracy of the results; • state how the calculation of sampling error is affected by imputation for nonresponse, misclassifications and	Not relevant since no sampling for enterprises with 10 and more employees is used.

			other sources of uncertainty, such as outlier treatment. If non-probability sampling is used, provide an assessment of representativity and risk of sampling bias. **ESS level** If probability sampling is used: • present sampling errors for key estimates across countries; • indicate which country to country differences are significant and which are not; • for a repetitive survey, describe at least broadly the trends in sampling error over time; • provide sampling errors for ESS level estimates.	
6.3	Non-sampling error	Error in estimates which cannot be attributed to sampling fluctuations	Summarise the most important aspects of coverage, measurement, non-response, processing and model assumption errors. Discuss the corresponding bias risks and actions undertaken to reduce them.	Some websites are not updated regularly. Selected websites have limited words describing the enterprise and may affect the final results. Not all enterprises have a website but they still can do the e-commerce or can be present on social media.
	A4. Unit non-response - rate (U)	The ratio of the number of units with no information or not usable information to the total number of in-scope (eligible) units, at a level of detail appropriate for a user report.		Not response rate is 0,3% for URLs and 0,5% for OBECs data.

	A5. Item non-response - rate (U)	The ratio of the in-scope (eligible) units that have not provided a particular item and the in-scope units that are expected to provide that particular item, at a level of detail appropriate for a user report.		Not relevant.
6.3.1	Coverage error	Divergence between the population of the Big Data source and the target population.	Provide information on the frame and its sources and actions performed to gather the population impacting on coverage (e.g. webscraping). Provide an assessment, whenever possible quantitative, of overcoverage and undercoverage, including an evaluation of the bias risks associated with the latter. Describe actions taken for reduction of undercoverage and associated bias risks.	For URLs Inventory there is no divergence between the population of big data source and *ICT usage in enterprise survey* target population. For OBECs, there is no divergence between sub-population of eneterpises having website and their websites. There will be over-coverage of statistical units having websites and no data for enterprises not having websites regarding OBEC indicator – social media presence.
6.3.1.1	A2. Overcoverage – rate (P)	The proportion of units accessible via the frame that do not belong to the target population.	Report A2, Overcoverage - rate	Not relevant.
6.3.2	Measurement error	Measurement errors are errors that occur during data capture and cause recorded values of variables to be different from the true ones	The main sources of measurement error should be reported and assessed. Their description should be accompanied by any available analysis, otherwise by the producer's best knowledge. Where available and relevant describe: • identification and general assessment of the main sources of measurement error, including errors arising from data acquisition;	The various ways to scrape and extract the text from websites could be source for measurement errors. The discrepancy between the lists of key words and the content of webpages could cause also a measurement error regarding OBECs data.

			• efforts made in questionnaire design and testing, information on interviewer training and other work on error prevention; • errors in measurement instruments (meters, satellites,...); • results of assessments based on comparisons with external data, re-interviews or experiments; • results of indirect analysis, for example, of the editing phase; and • actions taken to correct measurement errors.	
6.3.3	Non response error	Nonresponse errors occur when the Big Data source fails to collect one or all the variables for units belonging to the domain covered by the source	Provide qualitative/quantitative assessments of unit nonresponse and highlight the units that are most subject to nonresponse. Highlight the variables that are most subject to item nonresponse. Provide a qualitative/quantitative assessments of the bias associated with nonresponse, comparing response rate for different sub-groups or distribution of auxiliary variables known for respondents and non-respondets (etc.). Provide a breakdown of nonrespondents according to cause for nonresponse mainly focusing on unit dependent cause and data collection tools cause. Define a stategy for reducing nonresponse during data collection and follow-up. Implement an estimator adjusted for nonresponse.	There are some cases in which the website is not responding, blocking robots or they are in maintenance mode which can cause non response errors (Not response rate is 0,3% for URLs and 0,5% for OBECs data.).

			European level: Provide a qualitative/quantitative assessments of unit and item nonresponse across countries.	
6.3.3.1	A4. Unit nonresponse - rate (P)	The ratio of the number of units with no information or not usable information to the total number of in-scope (eligible) units, at a level of detail appropriate for a producer report.	Report A4: Unit nonresponse rate overall and at a level of detail appropriate for a producer report.	Not response rate is 0,3% for URLs and 0,5% for OBECs data.
6.3.4	Processing error	The error in final data collection process results arising from the faulty implementation of correctly planned implementation methods, e.g., algorithms used to transform the data or extract information from raw data.	If processing errors are significant, identify the main issues regarding them. Present an analysis of processing errors, where available, otherwise a qualitative assessment. Report their extent, and impact on the outputs, of the most significant types of error. Include descriptions of linking and coding errors, if applicable. Where mistakes relating to programming or publishing have occurred, corrective measures taken as well as actions for avoiding them in the future should be reported. Example: For web data sources: Setting up a pipeline assures processing is comparable over time. Because texts were processed, the final results were highly affected by the various choices of text processing made.	The processing of OBECs data is in three-steps: 1) create a document term matrix; 2) apply a Logistics regression model and 3) apply a deterministic check on the final results. Since the intermediate results from the ML procedure are merged together and then processed manually – the final accuracy between predicted and final results can be reported (is 68%).

6.3.4.1	A7. Imputation – rate	The ratio of the number of replaced values to the total number of values for a given variable.	Provide values of indicator A7 Imputation – rate.	Not applicable.
6.3.5	Model assumption error	Error due to models used in the statistical production.	Describe process specific models, for example, as needed to define the target of estimation itself and models used for transformation of data into statistical data. Provide an assessment of the validity of each model. Descriptions of models used in treatment of specific sources of error should be presented in the section dealing with those errors. The assessment of the models used in treatment of specific sources of error should be presented in this section. Discuss the trade off between the need to use proper model that can change over time (accuracy) and the use a constant model in order to ensure comparability over time	In ML approach there are different quality measures, such as accuracy. For URLs the accuracy is 92%, for OBECs is around 89%. It was not easy to find errors in deterministic approach.
6.5	Data revision - policy	Policy aimed at ensuring the transparency of disseminated data, whereby preliminary data are compiled that are later revised.	Describe the data revision policy applicable to data output from the statistical process being reported. In so far as they are relevant to the process being reported, summarise the general procedures for treatment of planned revisions, benchmark revisions, unplanned revisions, and revisions due to conceptual and/or methodological changes.	Not relevant.

			European level Describe the data revision policy and procedures at European level.	
7	**Timeliness and punctuality**	(Defined by its sub-concepts)	(Information relating to this concept is provided by reporting on its sub-concepts.)	
7.1	Timeliness	Length of time between data availability and the event or phenomenon the data describe.	Outline the reasons for the time lag. Outline efforts to reduce time lag in future. Describe the envisioned time lag for producing statistical output from/with the help of a new data source. Describe if the use of the new data source has the potential to decrease the time lag as it exists at the moment for already existing statistical products.	The results are avaialble after web scraping and processing information - it means that there is significant reduce in time needed to deliver the results. For the Bulgarian OBECs statistics it takes four weeks for the single process. It means that the experimental statistics for OBECs could be produce each month.
8	**Coherence and comparability**	Adequacy of statistics to be reliably combined in different ways and for various uses and the extent to which differences between statistics can be attributed to differences between the true values of the statistical characteristics.	(Information relating to this concept is provided by reporting on its sub-concepts.)	

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8.1	Comparability - geographical	The extent to which statistics are comparable between geographical areas.	Describe any problems of comparability between regions of the country. The reasons for the problems should be described and as well an assessment (preferably quantitative) of the possible effect on the output values. Give information on discrepancies from the ESS/ international concepts, definitions, with reference to other chapters for more details.	Each statistical unit (enterprise) is mapped to Statistical Business Register and NUTS classification.
8.2	Comparability - over time	The extent to which statistics are comparable or reconcilable over time.	Provide information on possible limitations in the use of data for comparisons over time. Distinguish three broad possibilities: 1. There have been no changes, in which case this should be reported. 2. There have been some changes but not enough to warrant the designation of a break in series. 3. There have been sufficient changes to warrant the designation of a break in series. Additionally, provide information about the comparability over time of the technological processes which produce the data, of the data access and changes in the covered population over time. Give also an assessment how the the comparison over time will develop in the future.	No any limitations in the use of OBECs data for comparisons over time.

8.3	Coherence - cross domain	The extent to which statistics are reconcilable with those obtained through other data sources or statistical domains.	An analysis of incoherence should be provided, where this is an issue of importance.	The *ICT usage in enterprises survey* data can be reproduced for some indicators.
8.4	Coherence - internal	The extent to which statistics are consistent within a given data set.	Each set of outputs should be internally consistent. If statistical outputs within the data set in question are not consistent, any resulting lack of coherence in the output of the statistical process itself should be stated as well as a brief explanation of the reasons for publishing such results.	Not applicable.
8.A.1	Coherence - with existing information/ Official Statistics	The extent to which information / statistical output from new data sources is consistent with information /statistical output from traditional data sources.	Provide information if it is meaningful to compare the information gained from new data sources with information from traditional data sources and if so, how consistent the information /statistical output gained from new data sources is with the one from traditional data sources.	The OBECs experimental indicators can be compared with results from ICT traditional survey, excluding the social media presence indicator with respect to the over-coverage of enterprises having website.
8.A.2	Comparability - between information from several distinct new data sources	The extent to which information from several distinct new data sources is comparable among one another.	If you have raw data from several distinct new data sources, provide information how comparable the respective raw data sets and the information derived from them are among one other. Examples: MNO data from several mobile operators, smart meter data from several electricity providers	Only one new data source – enterprise's websites - were analyzed.

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9	**Accessibility and clarity**	The conditions and modalities by which users can access, use and interpret data.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
9.6	Documentation on methodology	Descriptive text and references to methodological documents available.	List national reference metadata files, methodological papers, summary documents and handbooks relevant to the statistical process. For each item provide the title, publisher, year and link to on-line version (if any). List deliverables, reference metadata files, methodological papers, summary documents etc relevant to the process of deriving statistical data from raw data and - if already available - for producing statistical output using the statistical data.	Reference Methodological Framework, ver. 2.0, available on the following link: RMF 2.0.
9.7	Quality documentation	Documentation on procedures applied for quality management and quality assessment.	List relevant quality related documents, for example, other quality reports, studies. Cross reference to descriptions of quality procedures in other chapters, especially concept 6. List also the deliverables, in which quality related issues are described.	Reference Methodological Framework, ver. 2.0, available on the following link: RMF 2.0. For quality procedures, please see concept 6.

10	**Cost and burden**	Cost associated with the collection and production of a statistical product and burden on respondents.	Cost Provide annual operational costs of the process, with breakdown by major cost component. Describe recent efforts to improve efficiency and comment on the extent to which information and communication technology is used. European level Describe recent initiatives and efforts to improve efficiency at the European level. Burden Provide an estimate of the respondent burden imposed by the process. Describe all the means taken to minimise burden. **European level** Describe recent initiatives and efforts to minimise burden at the European level.	The search engine is queried every 7 seconds and enerprise's websites every 0,5 seconds to not overload webservers. The time consuming of the experts can be consider.
10.A	Potential savings in cost and burden	Description how the new data source might influence cost and burden in the future	Provide an overview how the new data source could be deployed in the future to save the NSIs cost and/or decrease the respondent burden. Provide a qualitative description of the additional efforts for the NSI and the data owners.	Only few questions from *ICT usage in enterprises survey* can be supported with the approach. However, new OBECs data of enterprises, not covered in traditional survey, can be provided.

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11	**Confidentiality**	A property of data indicating the extent to which their unauthorised disclosure could be prejudicial or harmful to the interest of the source or other relevant parties.	(Information relating to this concept is provided by reporting on its sub-concepts.)	
11.1	Confidentiality - policy	Legislative measures or other formal procedures which prevent unauthorised disclosure of data that identify a person or economic entity either directly or indirectly.	Describe all European or national legislation, or other formal requirements, that relate to confidentiality. Describe relevant policy (if any). Note that the existence of legislation and/or policy provides some assurance that methods necessary to assure confidentiality have been applied to the data. European level Summarise the commonalties and differences in national approaches to confidentiality policy.	Websites have public access so no restrictions regarding confidentiality are applied. The ESS web-scraping policy template (Deliverable C1) was followed.
11.2	Confidentiality - data treatment	Rules applied for treating the datasets to ensure statistical confidentiality and prevent unauthorised disclosure.	For aggregate outputs • Provide the rules that define a confidential cell. • Describe the procedures for detecting confidential cells, including checking for residual disclosure. • Describe the procedures for eliminating confidential cells, for example by controlled rounding, cell suppression, or cell aggregation. For micro-level outputs: • Describe the procedures that are used in protecting confidentiality.	Not applicable.

11.A1	Privacy	How privacy sensitive is the information coming from external data holders?	State which treatments are prescribed to satisfy privacy concerns	Not applicable.
11.A2	Privacy - protecting treatments	Treatment applied to ensure privacy - sensitive information from external data holders	State any treatments prescribed to satisfy privacy concerns.	Not applicable.
12	**Comment**	Supplementary descriptive text which can be attached to data or metadata.	Provide any information that is pertinent to the report but does not fit under any of the other concepts, or to repeat key issues, or to make reference to annexes that might be attached to the report.	The data collection is carried March-April 2020 in order to be compare with ICT survey collection period.